Synthetic Native Template

Redistributable test fixture — not laboratory data

Representative content layout

- Native title/content placeholders

Progress / Previous Commitments

Current position | H002
Commitment | NS101 | done | Owner: Gary | Due: 2026-09-10T09:00:00Z
Dependencies | pressure fixture | Parallel: True
Commitment | NS201 | planned | Owner: Gary | Due: 2026-09-24T00:00:00Z
Dependencies | pressure fixture | Parallel: True

H001 ｜ Hypothesis

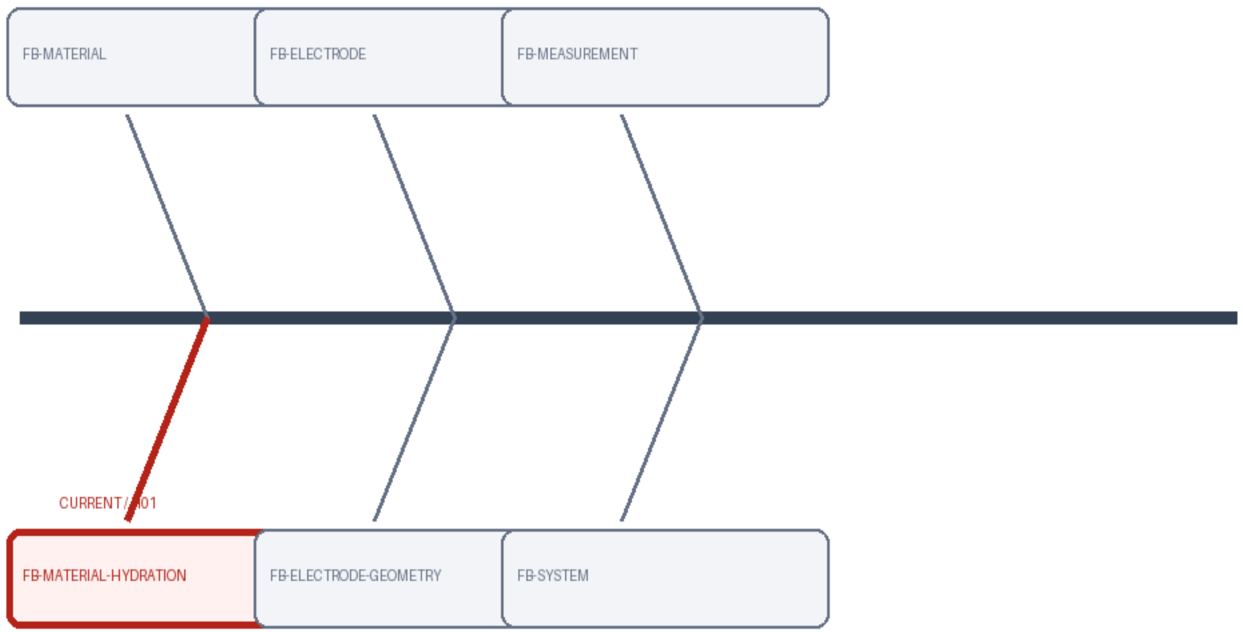
Hypothesis ｜ Bulk conductivity drives the positional signal gradient.
Falsifier ｜ Matched conditions do not change the predicted CV or resistance.
Research question ｜提升 bulk conductivity 是否足以降低位置依賴的訊號變異？

H001 ｜ Problem

Problem ｜合成薄膜的缺陷密度沿位置增加，且訊號重複性下降。
Previous finding ｜含水量與導電度存在位置梯度
Conflict ｜導電度提高是否足以修復重複性仍未知。
Research question ｜提升 bulk conductivity 是否足以降低位置依賴的訊號變異？
Scope ｜合成五個位置、每個位置三個 replicate 的機制辨識。

H001 | Total Fishbone / Research Map

Historical snapshot | FB001 rev1
Focus branch | FB-MATERIAL-HYDRATION
Immutable map; current branch highlighted.



H001 ｜ Observation → Literature → Mechanism

Observation ｜位置依賴缺陷與訊號變異已觀察到。
Problem ｜現有模型無法解釋此變異。
Research question ｜提升 bulk conductivity 是否足以降低位置依賴的訊號變異？
Literature consensus ｜ transport gradient 可產生位置效應。
Alternatives ｜ interface contact remains alternative
Gap ｜缺少控制比較隔離機制。
Implication ｜需匹配條件後比較。
Mechanism ｜ bulk transport
Strategy ｜均化導電度

H001 ｜ Literature → Mechanism → Strategy

Observation ｜位置依賴缺陷與訊號變異已觀察到。
Problem ｜現有模型無法解釋此變異。
Research question ｜提升 bulk conductivity 是否足以降低位置依賴的訊號變異？
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Implication ｜需匹配條件後比較。
Mechanism ｜ bulk transport
Strategy ｜均化導電度

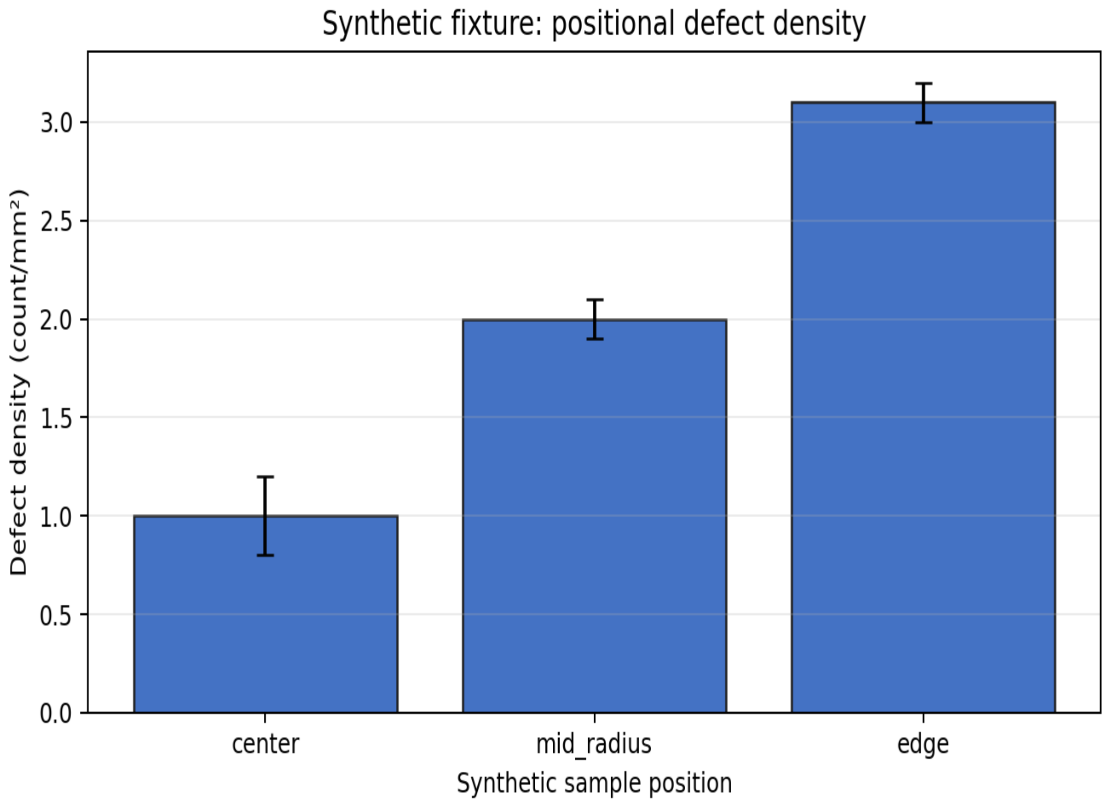
H001 ｜ Experiment Design

EXP101 ｜ IV: [' 含水量 ']
Controls: [' 原始配方 ']
N/replicates: {'replicates': 3, 'samples': 15}
Metrics/units: [' 導電度 (mS/cm)']
Method: ['synthetic-four-probe']
Prediction: [' 導電度提升 ']
Decision rule: {'go': 'mean conductivity increases >=20%', 'partial_go': '10-20%', 'no_go': '<10%'}

EXP102 ｜ IV: [' 位置 ']
Controls: [' 原始位置分布 ']
N/replicates: {'replicates': 3, 'samples': 15}
Metrics/units: [' 訊號 CV (%)']
Method: ['synthetic-signal-test']
Prediction: ['CV 下降 ']
Decision rule: {'go': 'CV decreases >=30%', 'partial_go': '10-30%', 'no_go': '<10%'}

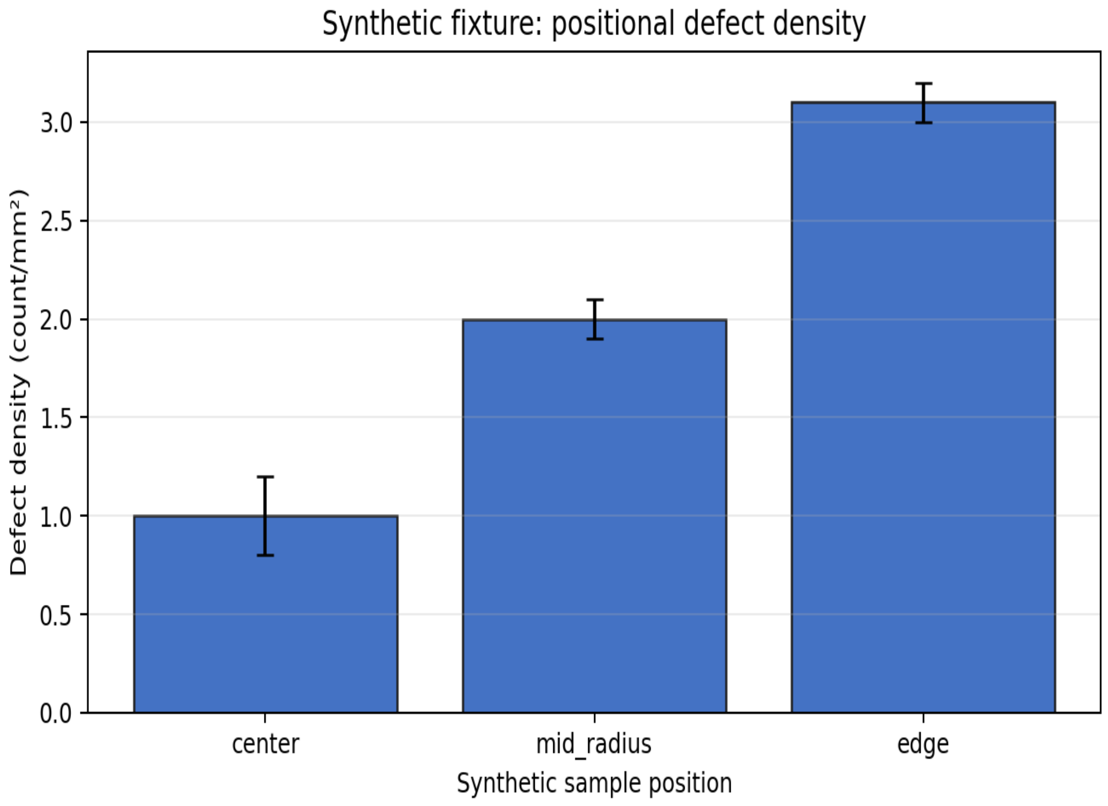
H001 ｜ Results

RES101 ｜平均導電度增加 24% ± 5% SD



H001 ｜ Results

RES102 ｜訊號 CV 僅下降 4% ± 6% SD ，屬 No-Go



H001 ｜ Integrated Discussion

Supporting ｜ RES101
Contradicting ｜ RES102
Cross-experiment pattern ｜導電度提升成立，但重複性沒有同步改善。
Mechanism assessment ｜ bulk conductivity 是部分機制，不足以解釋 interface-sensitive variation 。
Alternatives ｜接觸電阻；局部壓力差；電極幾何
Remaining uncertainty ｜ contact interface 是否為主導因素仍需匹配導電度後檢驗。

H001 ｜ Layer Summary / Decision

Answered ｜ bulk conductivity 影響缺陷梯度，但不足以單獨解釋重複性。

Hypothesis status ｜ partial_support

Decision ｜ Partial-Go: Bulk conductivity is contributory but insufficient.

Unresolved ｜ interface contact 的貢獻尚未量化。

Next question ｜匹配 bulk conductivity 後， contact resistance 是否主導變異？

Next Step ｜ NS101

H01 → H02 ｜ Hypothesis Transition

Previous hypothesis ｜ C101

Key results ｜ RES101, RES102

Not explained ｜導電度增加後訊號重複性沒有改善。

New observation ｜ E201

Therefore ｜剩餘變異隨電極接觸位置改變，因此核心解釋由 bulk transport 轉向 interface contact 。

New hypothesis ｜ C201

H002 ｜ Hypothesis

Hypothesis ｜ Contact resistance drives instability at low pressure.
Falsifier ｜ Matched conditions do not change the predicted CV or resistance.
Research question ｜低接觸壓力下， contact resistance 是否主導訊號不穩定？

H002 ｜ Problem

Problem ｜導電度改善後重複性仍未改善，剩餘變異與電極接觸位置相關。

Previous finding ｜ H01 支持導電度梯度存在 ; H01 未支持導電度單獨主導重複性

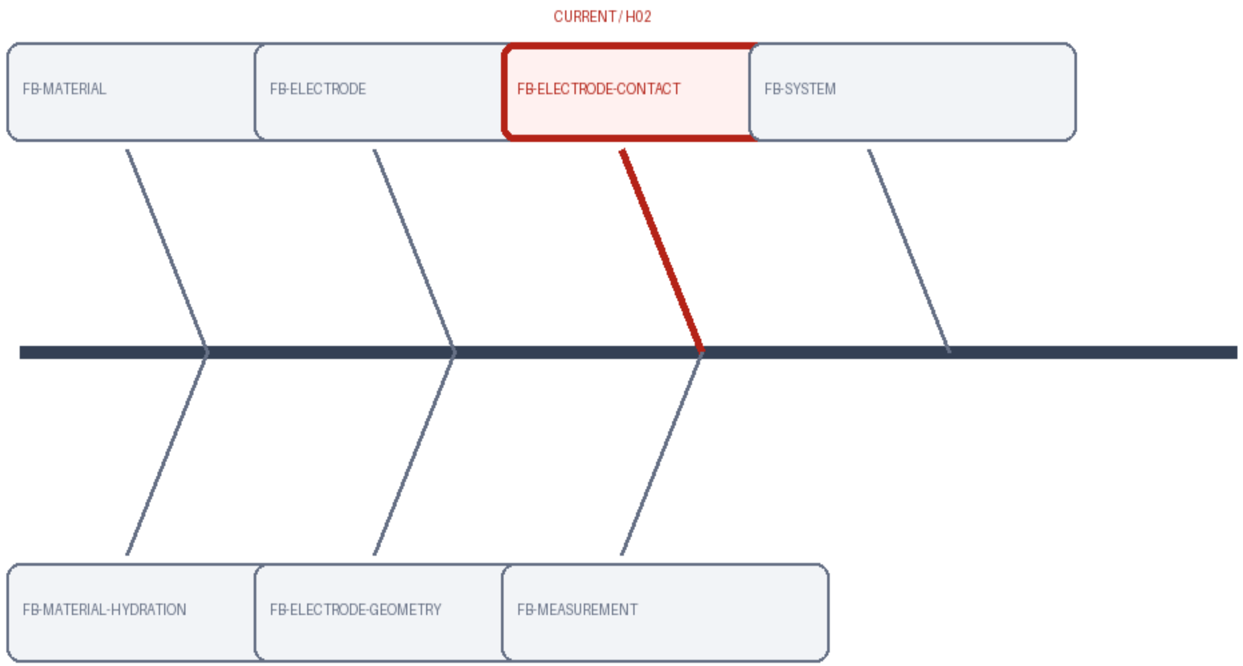
Conflict ｜ bulk transport 與 interface contact 的相對貢獻尚未區分。

Research question ｜低接觸壓力下， contact resistance 是否主導訊號不穩定？

Scope ｜匹配 bulk conductivity ，改變 contact pressure 的對照實驗。

H002 │ Total Fishbone / Research Map

Historical snapshot │ FB001 rev2
Focus branch │ FB-ELECTRODE-CONTACT
Immutable map; current branch highlighted.

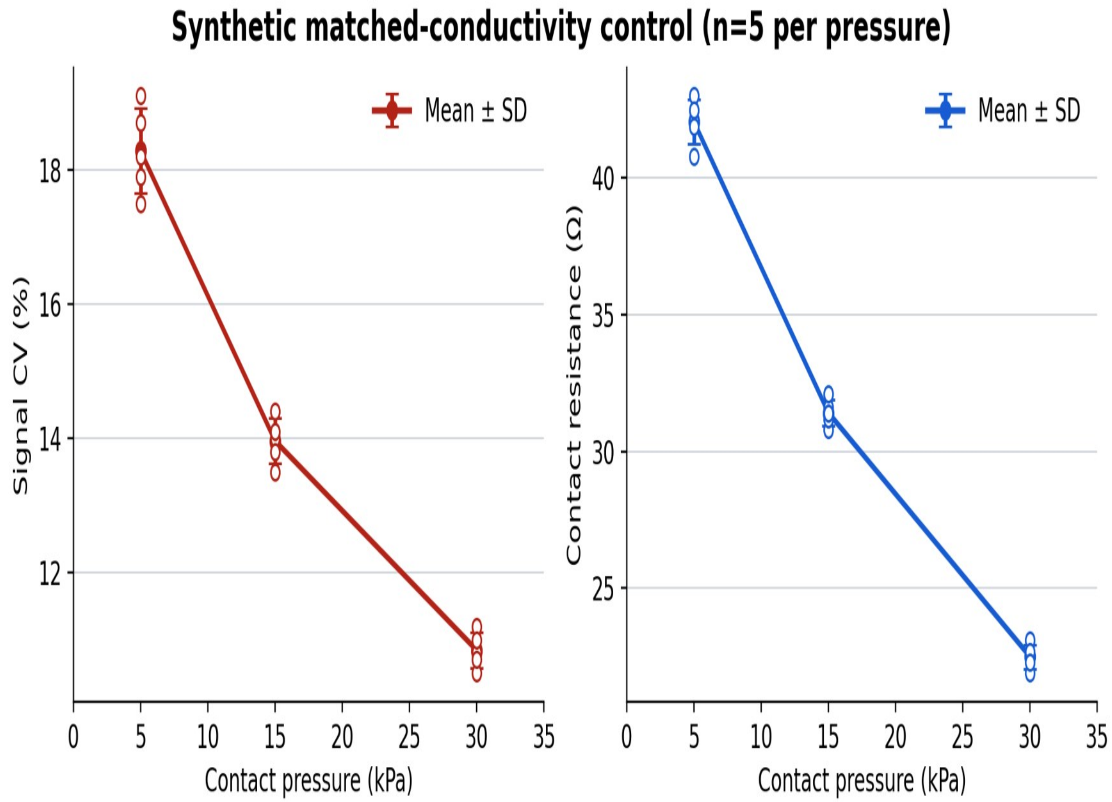


H002 │ Observation → Literature → Mechanism

Observation │位置依賴缺陷與訊號變異已觀察到。
Problem │現有模型無法解釋此變異。
Research question │低接觸壓力下， contact resistance 是否主導訊號不穩定？
Literature consensus │ transport gradient 可產生位置效應。
Alternatives │ interface contact remains alternative
Gap │缺少控制比較隔離機制。
Implication │需匹配條件後比較。
Mechanism │ contact resistance
Strategy │匹配導電度並改變接觸壓力

H002 ｜ Results

EXP201 ｜
RES201 ｜高壓條件 CV 下降 38% ± 7% SD ，contact resistance 同步下降



H002 ｜ Layer Summary / Decision

Answered ｜低接觸壓力下 contact resistance 是主導不穩定來源。

Hypothesis status ｜ supported

Decision ｜ Go: Contact-pressure control discriminates the interface mechanism.

Unresolved ｜高壓循環的耐久性與 wear 尚未測試。

Next question ｜接觸壓力改善能否在長期循環下維持？

Next Step ｜ NS201